Kernix 0.11.1 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

46 workloads

153 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

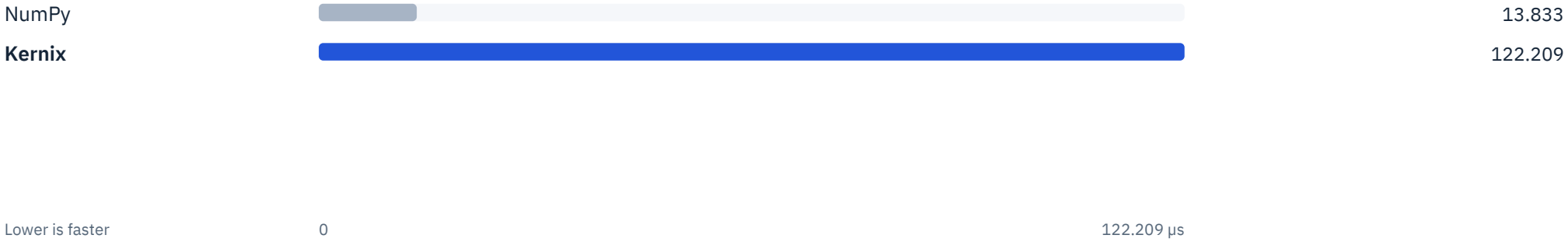
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication · transposed view

Per-call median

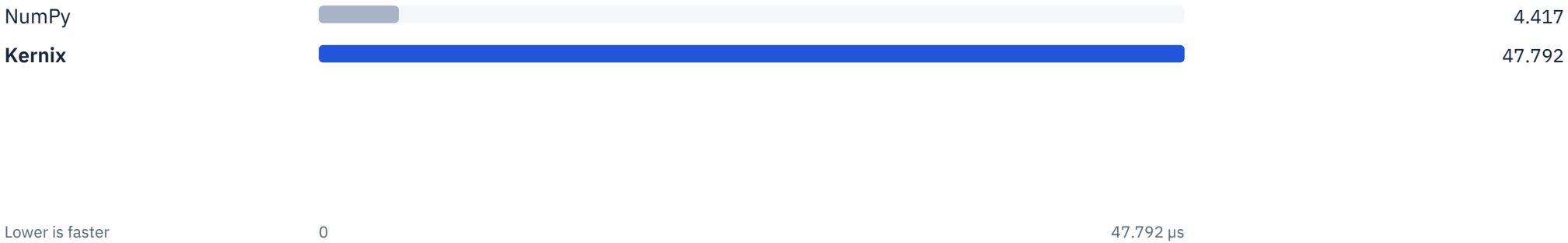
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



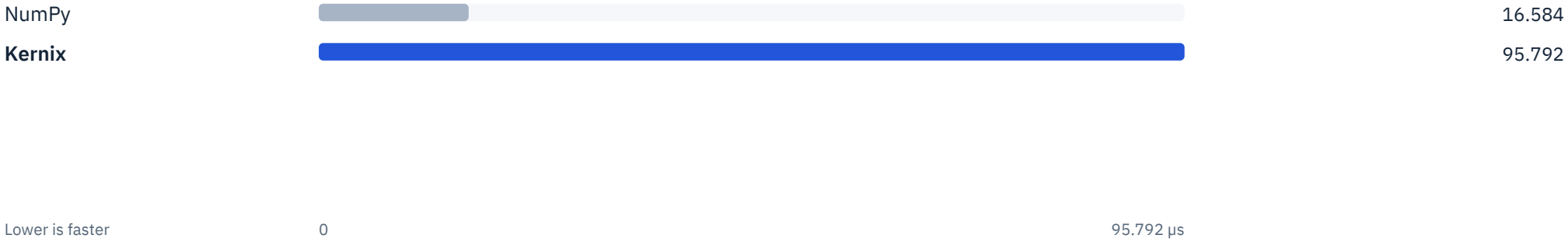
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

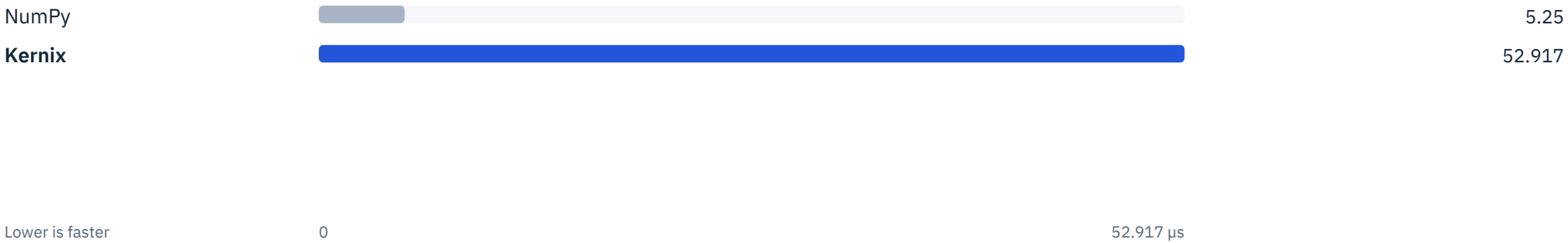
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



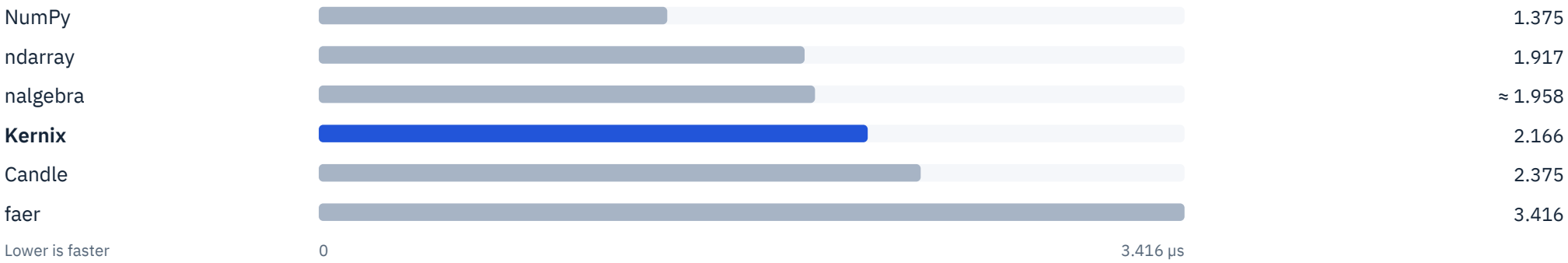
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

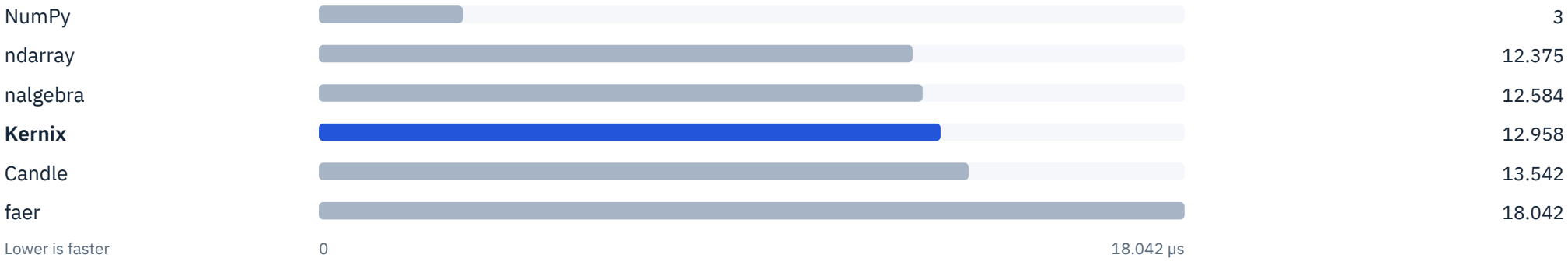
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



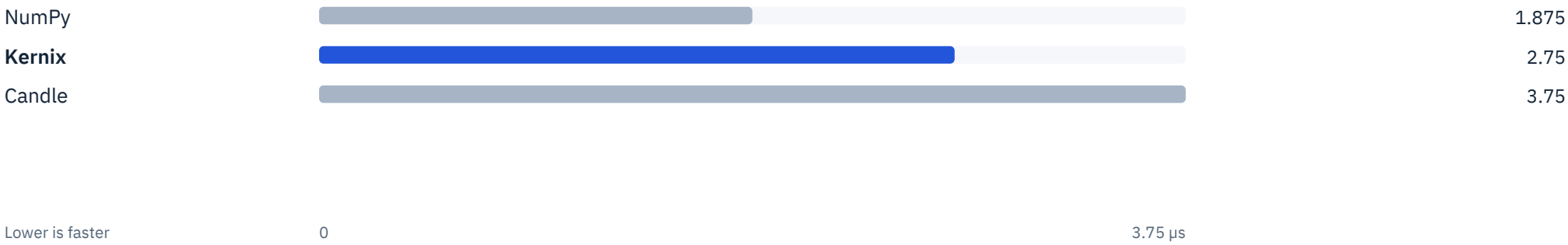
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

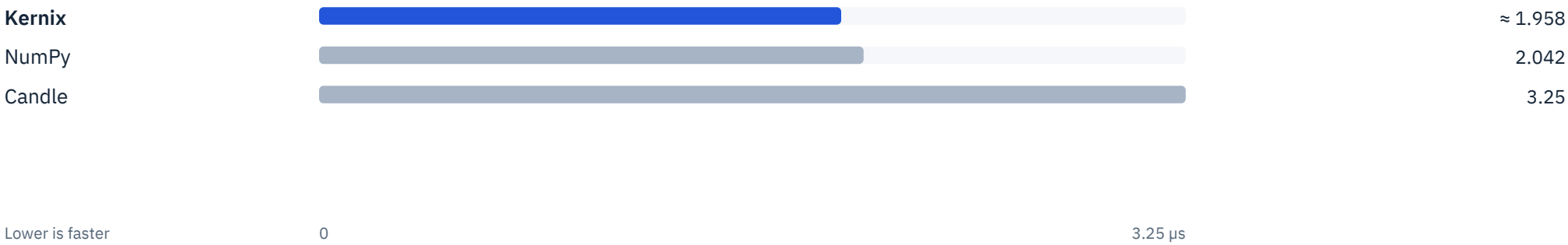
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



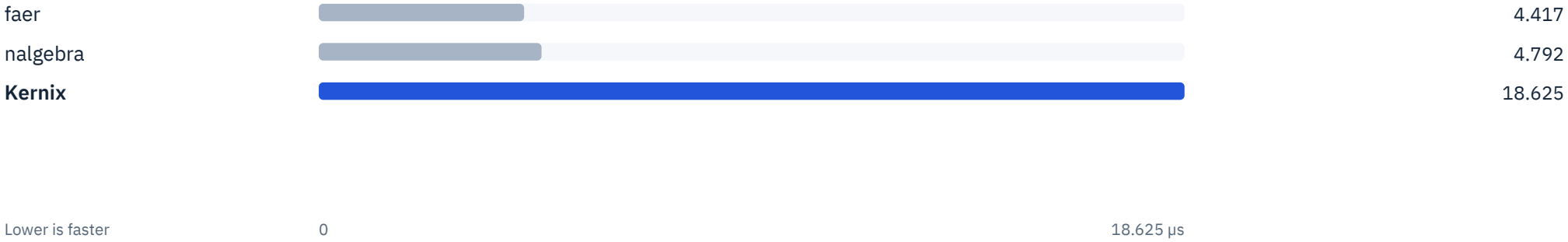
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

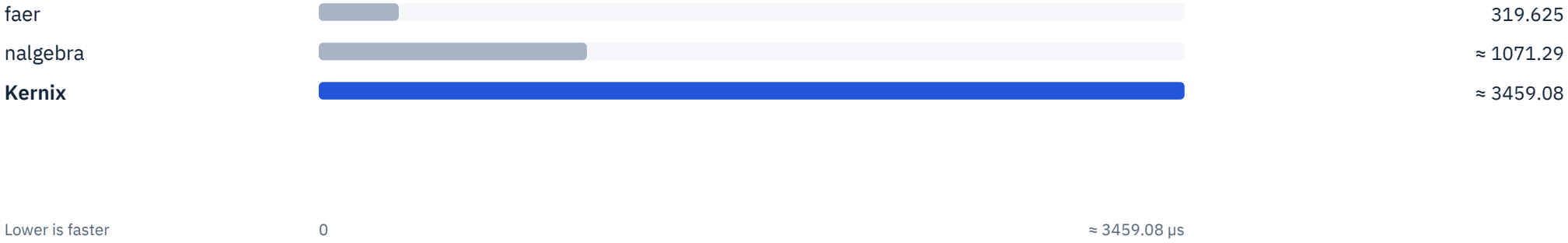
128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



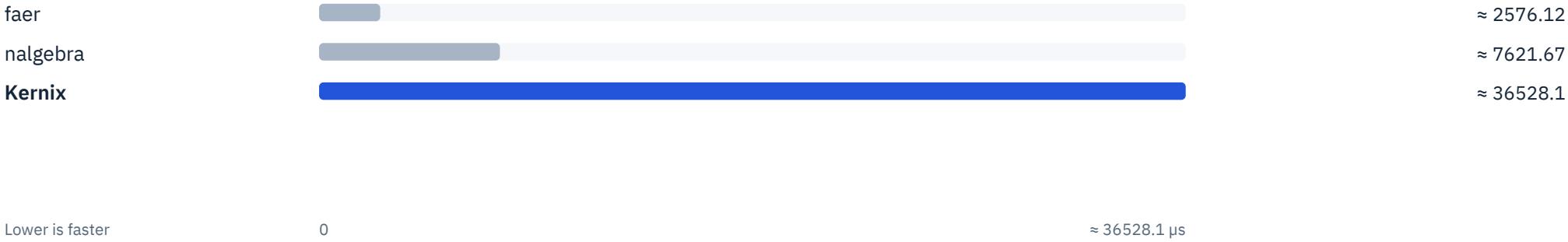
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky solve · reused factor

Per-call median

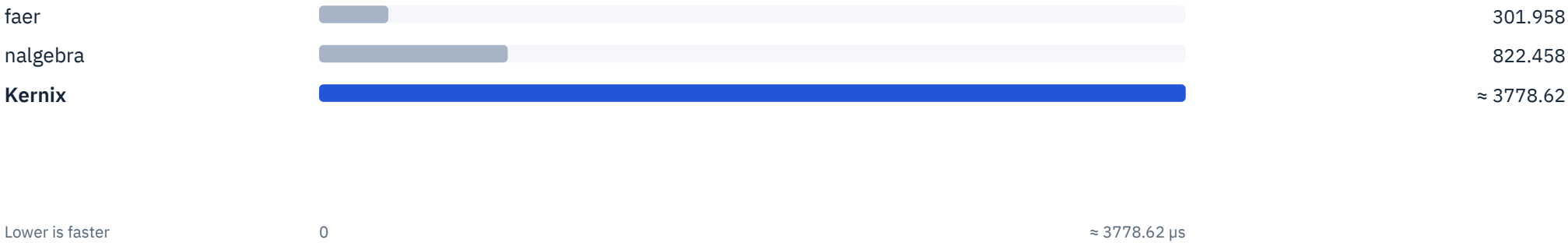
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



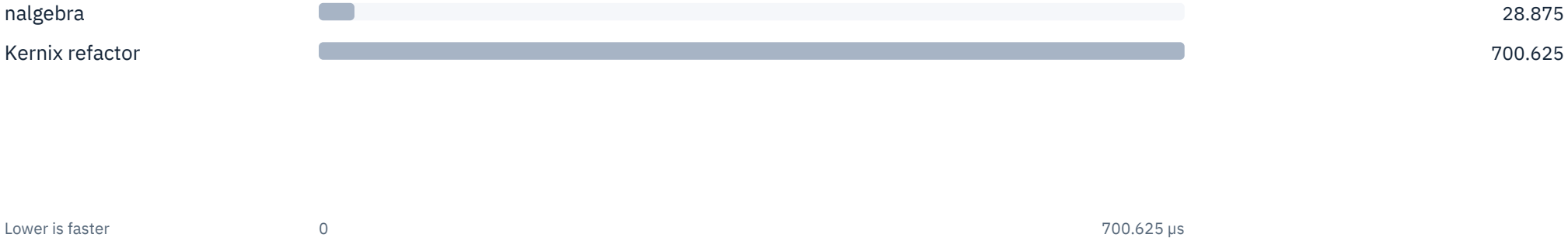
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

Per-call median

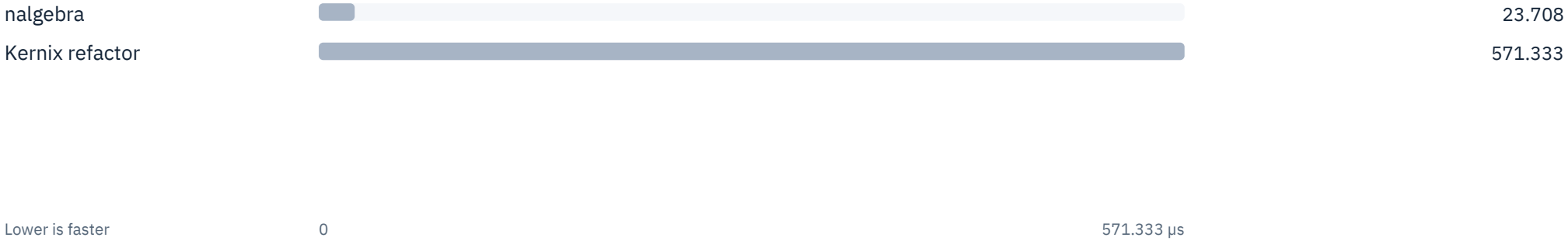
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output



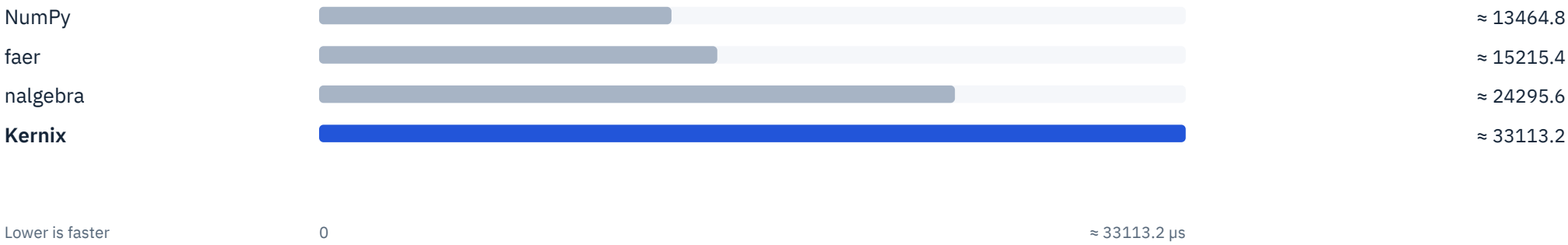
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

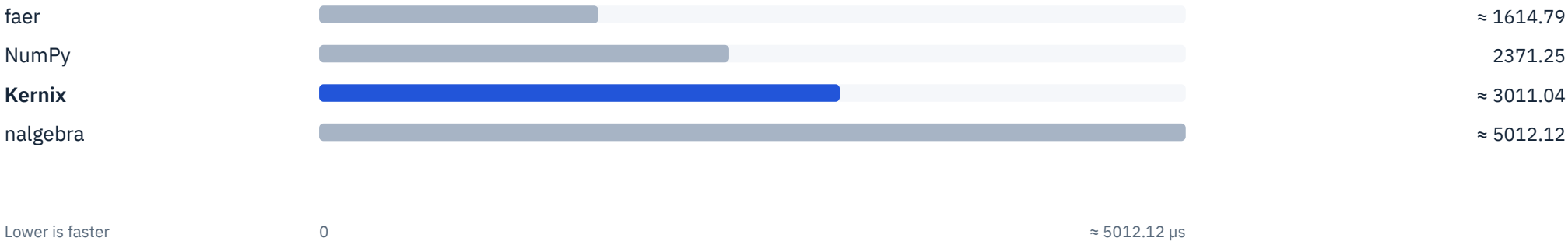
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



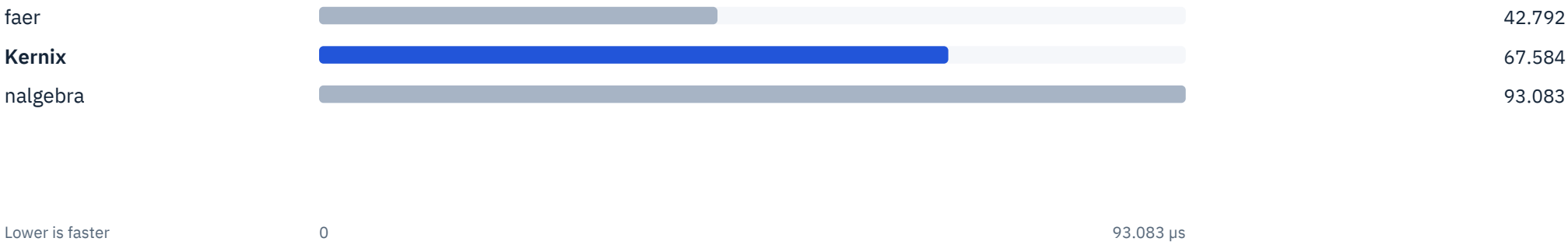
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

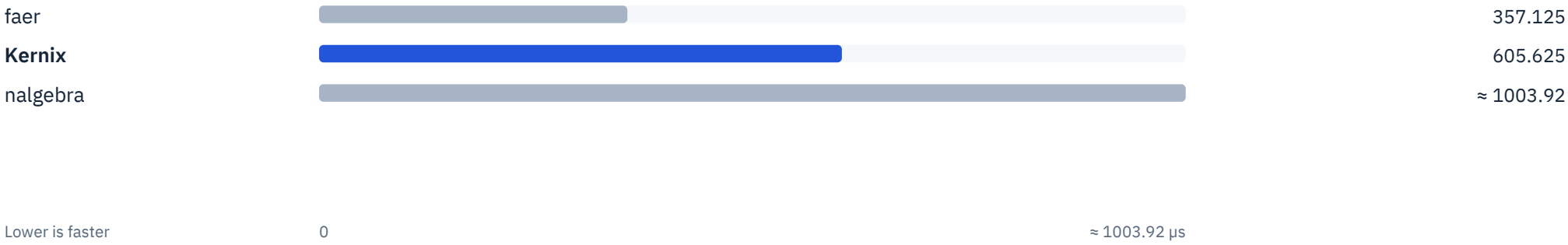
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



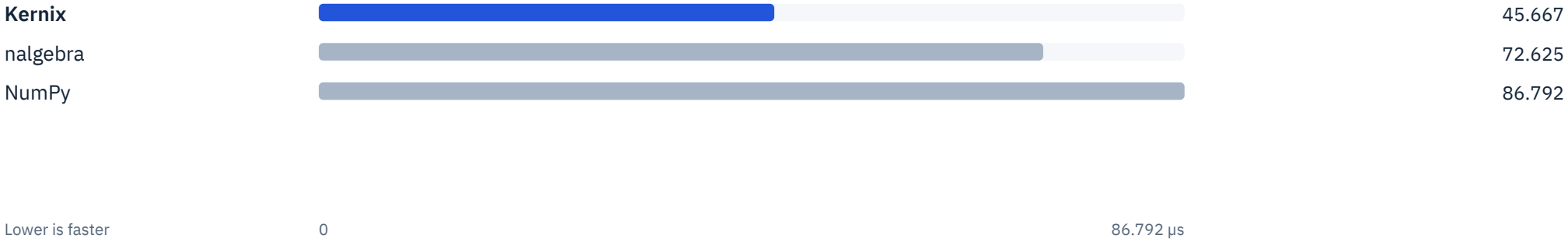
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

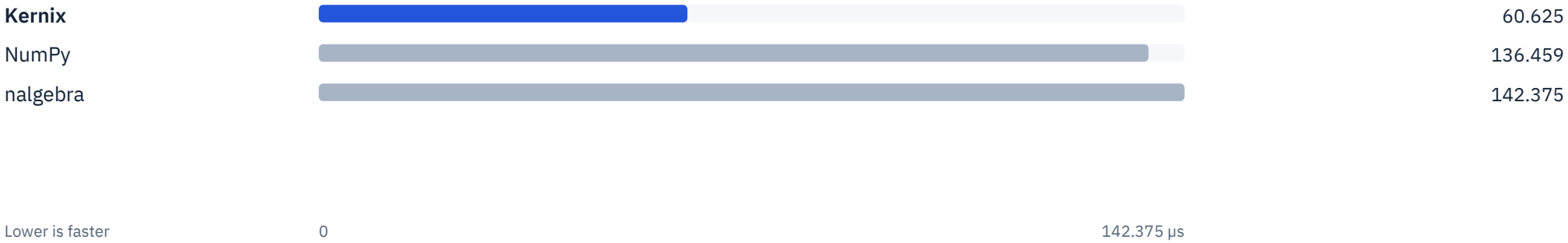
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

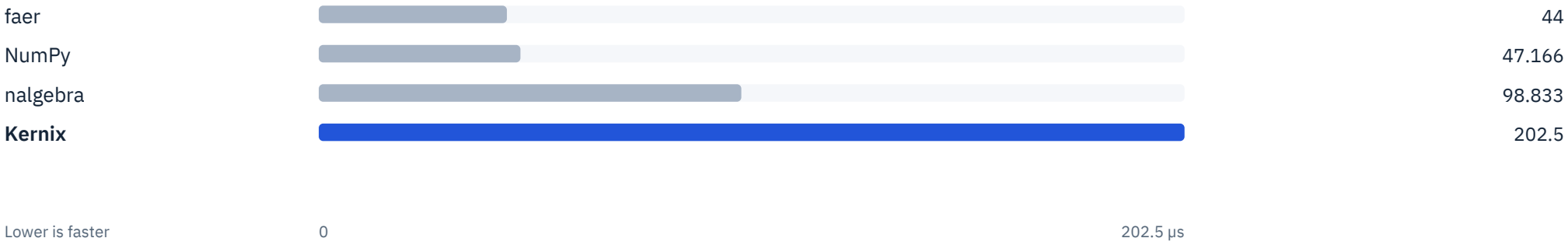
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

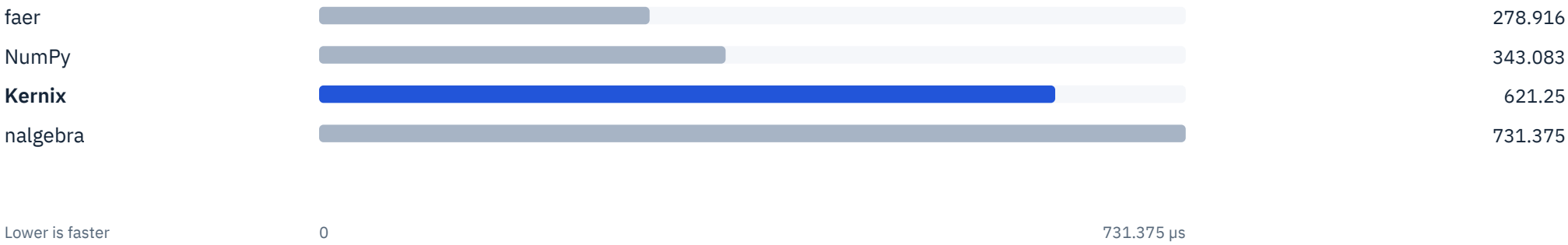


Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



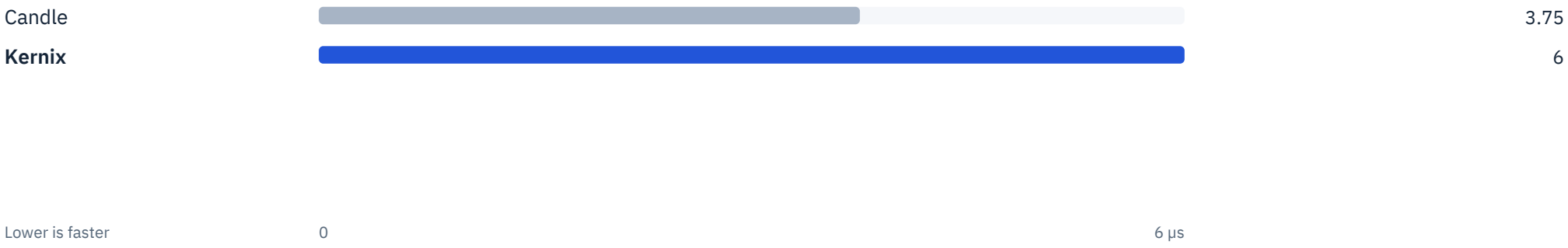
Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

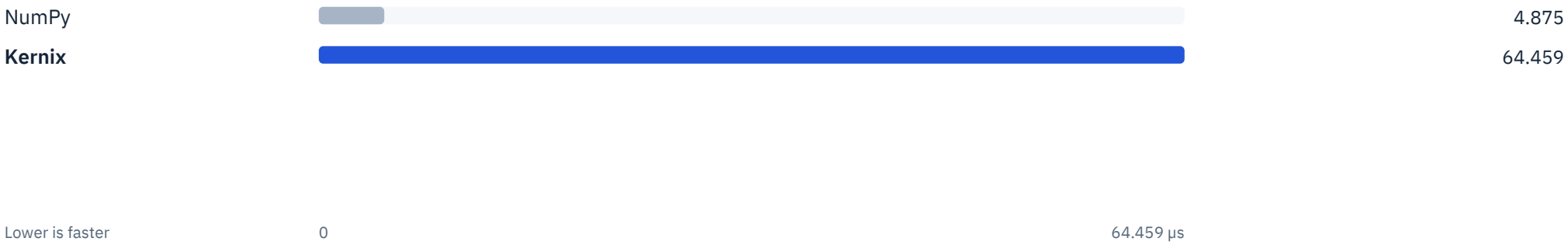


Kernix 0.11.1 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



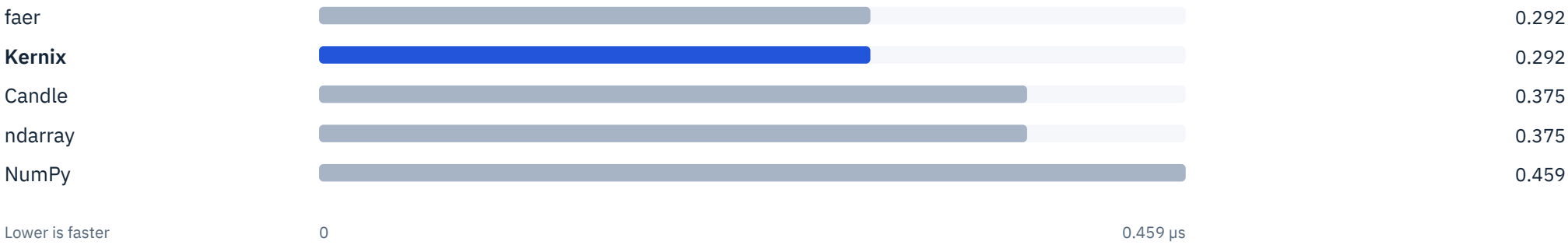
Kernix 0.11.1 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output

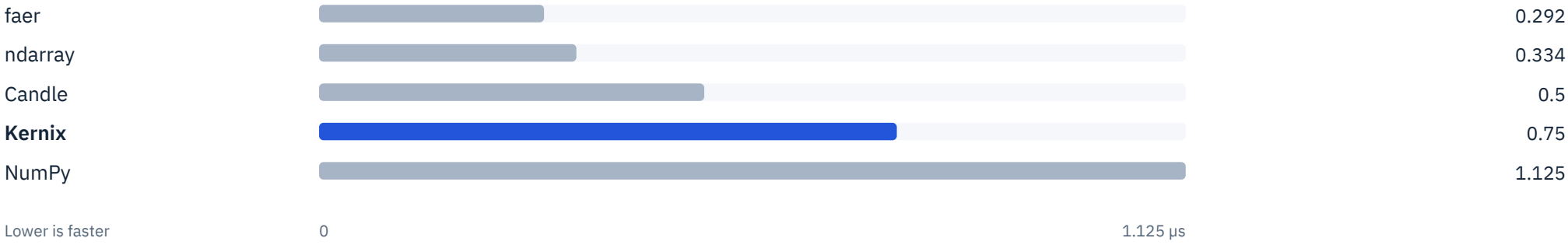
Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	13833	13.833
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	122209	122.209
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	4417	4.417
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	47792	47.792
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	95792	95.792
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	16584	16.584
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	5250	5.25
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	52917	52.917
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2375	2.375
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1958.0000000000002	1.9580000000000002

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2166	2.166
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1917	1.917
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1375	1.375
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3416	3.416
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12958	12.958
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	12584	12.584
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	12375	12.375
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3000	3
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	13542	13.542
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	18042	18.042

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	94167	94.167
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	94959	94.959
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	96750	96.75
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	16082.999999999998	16.082999999999998
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	95625	95.625
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	15042	15.042
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	117708	117.708
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	746459	746.459
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	726291	726.291
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	737417	737.417

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	99333	99.333
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	891708	891.708
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	120083	120.083
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	812166	812.166
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7333	7.333
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6750	6.75
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8375	8.375
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1916	1.916
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7083	7.083

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	47750	47.75
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	51750	51.75
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	49292	49.292
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	47792	47.792
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	56959	56.959
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	6667	6.667
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6000	6
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	385292	385.292
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	362500	362.5
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	360500	360.5

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	393417	393.417
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	36125	36.125
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	405917	405.917
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	36500	36.5
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	1875	1.875
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2750	2.75
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3750	3.75
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2042	2.042
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1958.0000000000002	1.9580000000000002
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	3250	3.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	19500	19.5
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	65375	65.375
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	403792	403.792
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	46792	46.792
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	4417	4.417
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4792	4.792
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18625	18.625
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	3459083	3459.083
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1071292	1071.292
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	319625	319.625

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	36528084	36528.084
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2576125	2576.125
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7621667	7621.667
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	822458	822.458
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	3778625	3778.625
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	301958	301.958
Cholesky rank-one update / nalgebra 256 × 256 · f64	28875	28.875
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	700625	700.625
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	571333	571.333
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	23708	23.708

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / nalgebra 512 × 512 · f64	89667	89.667
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3465667	3465.667
Thin QR / nalgebra 512 × 128 · f64	2785125	2785.125
Thin QR / NumPy 512 × 128 · f64	2049125	2049.125
Thin QR / faer 512 × 128 · f64	1959125	1959.125
Thin QR / Kernix 512 × 128 · f64	4014292	4014.292
Thin QR / Kernix 1024 × 256 · f64	33113249.999999996	33113.249999999996
Thin QR / faer 1024 × 256 · f64	15215375	15215.375
Thin QR / NumPy 1024 × 256 · f64	13464750	13464.75
Thin QR / nalgebra 1024 × 256 · f64	24295583	24295.583

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	5012125	5012.125
Cholesky factorization / faer 512 × 512 · f64	1614792	1614.792
Cholesky factorization / Kernix 512 × 512 · f64	3011042	3011.042
Cholesky factorization / NumPy 512 × 512 · f64	2371250	2371.25
Column-pivoted QR / nalgebra 96 × 32 · f64	93083	93.083
Column-pivoted QR / faer 96 × 32 · f64	42792	42.792
Column-pivoted QR / Kernix 96 × 32 · f64	67584	67.584
Column-pivoted QR / nalgebra 256 × 64 · f64	1003917	1003.917
Column-pivoted QR / Kernix 256 × 64 · f64	605625	605.625
Column-pivoted QR / faer 256 × 64 · f64	357125	357.125

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	45667	45.667
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	72625	72.625
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	86792	86.792
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	142375	142.375
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	60625	60.625
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	136459	136.459
Thin QR / nalgebra 96 × 32 · f64	39209	39.209
Thin QR / Kernix 96 × 32 · f64	53042	53.042
Thin QR / faer 96 × 32 · f64	43000	43
Thin QR / NumPy 96 × 32 · f64	49125	49.125

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 256 × 64 · f64	369084	369.084
Thin QR / Kernix 256 × 64 · f64	516250	516.25
Thin QR / nalgebra 256 × 64 · f64	392875	392.875
Thin QR / faer 256 × 64 · f64	356750	356.75
Cholesky factorization / faer 64 × 64 · f64	9959	9.959
Cholesky factorization / Kernix 64 × 64 · f64	21917	21.917
Cholesky factorization / nalgebra 64 × 64 · f64	21500	21.5
Cholesky factorization / NumPy 64 × 64 · f64	13750	13.75
Cholesky factorization / faer 128 × 128 · f64	44000	44
Cholesky factorization / NumPy 128 × 128 · f64	47166	47.166

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 128 × 128 · f64	98833	98.833
Cholesky factorization / Kernix 128 × 128 · f64	202500	202.5
Cholesky factorization / nalgebra 256 × 256 · f64	731375	731.375
Cholesky factorization / NumPy 256 × 256 · f64	343083	343.083
Cholesky factorization / faer 256 × 256 · f64	278916	278.916
Cholesky factorization / Kernix 256 × 256 · f64	621250	621.25
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3750	3.75
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6000	6
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7125	7.125
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3083	3.083

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Kernix batched normalized/sliced MLP · f64	2167	2.167
Model inference / Candle batched normalized/sliced MLP · f64	5875	5.875
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3250	3.25
Model recording + forward / Candle batched normalized/sliced MLP · f64	6333	6.333
Model backward / Candle prepared graph → fresh gradients · f64	23375	23.375
Model backward / Kernix prepared graph → fresh gradients · f64	3250	3.25
Parameter update / Kernix transactional SGD, fixed gradients · f64	791	0.791
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	4875	4.875
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	64459	64.459
Row softmax / Kernix 256 × 64 · f64	84125	84.125

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	86959	86.959
Row softmax / Candle 256 × 64 · f64	74792	74.792
Row softmax / ndarray 256 × 64 · f64	71667	71.667
Create zeros / NumPy 64 × 64 · f32	459	0.459
Create zeros / Kernix 64 × 64 · f32	292	0.292
Create zeros / Candle 64 × 64 · f32	375	0.375
Create zeros / ndarray 64 × 64 · f32	375	0.375
Create zeros / faer 64 × 64 · f32	292	0.292
Create ones / faer 64 × 64 · f32	292	0.292
Create ones / NumPy 64 × 64 · f32	1125	1.125

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Kernix 64 × 64 · f32	750	0.75
Create ones / Candle 64 × 64 · f32	500	0.5
Create ones / ndarray 64 × 64 · f32	334	0.334